Ipotesi a confronto: priorità di ricerca, posizione del modello ed evidenza osservata

Nove ipotesi che coprono i due obiettivi di ricerca e diversi tipi di intervento, scelte per copertura e non per merito. Tre cose restano separate, in blocchi distinti: la priorità di ricerca (posizione F, C o P e motivo del piano di verifiche: dice dove portare il prossimo studio, non se il farmaco funziona), la posizione del modello e l'evidenza osservata in un sistema WWOX. Nessun punteggio combinato, nessun ordine.

Il testo scientifico dei blocchi è conservato in inglese, come nel confronto originale; potenza ed esposizione non sono riportate in questa versione. **Nessuna probabilità clinica calibrata è mostrata, perché nessuna è stata validata; il percentile del modello non è mostrato affatto.**

4-phenylbutyrate class and bile-acid chemical chaperones CMP-01

PRIORITÀ DI RICERCA

4-phenylbutyrate class and bile-acid chemical chaperones A01_pba_class

F3 sul bisogno Salvare una proteina WWOX destabilizzata da una variante missenso del dominio SDR

Piano di verifiche:

risolvere una premessa

Salvare una proteina WWOX destabilizzata da una variante missenso del dominio SDR

POSIZIONE DEL MODELLO

Il modello non si applica a questa ipotesi (non valutata)

EVIDENZA OSSERVATA IN UN SISTEMA WWOX

None for the intervention in any WWOX system. The lesion side: one patient-fibroblast blot (one membrane, one exposure) shows transcript comparable to reference lines and no detectable protein band; fractions not separated.

▼ Dettagli: contesto, evidenza favorevole, contraria e mancante, prossimo esperimento

Track e genotipo

Track B: lesion-specific, a Q230P-class missense with residual protein; no meaning for null, nonsense or splice lesions

Profondità del beneficio cercato

deep: function per molecule of the rescued protein; a larger amount of protein is not function and is not read as rescue

Favorevole: premessa dimostrata

Other proteins: chemical chaperone or osmolyte raised a destabilised mutant with protein and function read together in some cases (GAT-1 variant; a soluble cytosolic client roughly doubled in soluble fraction). All are outside WWOX.

Contraria: effetti opposti o controevidenza

Wild-type protein rises too (not mutant-specific); partly trafficked variant not rescued by the class in one precedent; butyrate raised immature total protein with small function gain; toxicity ceiling near 4 mM in one paper; HDAC inhibition is quantitatively available at the applied levels and could be the operating mechanism, with transcriptional side effects.

Mancante: premessa non dimostrata

Where the Q230P-class protein is lost (never made, cleared, insoluble, epitope-invisible); any per-molecule WWOX function assay; whether the lesion is a stability lesion; which of three mechanism classes (chaperoning, HDAC6/HSP90 acetylation, class-I HDAC transcription) operates.

Precedenti esterni (non sono evidenza WWOX)

Several cell, worm, fly, zebrafish and short mouse precedents in other proteins: cards record complete or partial full-text reads for the R1 papers; no WWOX test. Rationale was withdrawn once in this repository and narrowed, not refuted (IT5 P8).

Prossimo esperimento
Build the wild-type function ruler first: isogenic WWOX-null human neural line, wild-type WWOX titrated over at least five levels with a Tier 1 WW1-dependent readout and measured protein; then place the Q230P-class allele on the curve; only then a lever panel (class-I HDAC inhibitor; HDAC6-selective or pan-HDAC; HDAC-free chaperone and osmolytes) with mRNA flat and the null line unaffected as conditions. Outcomes fixed in advance in the card.

Stato dell'esperimento
PROPOSED, not run; no measurement of this kind exists in the repository; its absence does not block the computational work

Proteasome-inhibitor class CMP-02

PRIORITÀ DI RICERCA

Proteasome-inhibitor class
A02_proteasome_inhibitor_class
F3 sul bisogno Salvare una proteina WWOX destabilizzata da una variante missenso del dominio SDR

Piano di verifiche:
risolvere una premessa
Salvare una proteina WWOX destabilizzata da una variante missenso del dominio SDR

POSIZIONE DEL MODELLO

il modello non si applica a questa ipotesi (non valutata)

EVIDENZA OSSERVATA IN UN SISTEMA WWOX

None for the intervention. No fractionation, half-life, inhibitor or synthesis datum exists for the exact variant.

▼ Dettagli: contesto, evidenza favorevole, contraria e mancante, prossimo esperimento

Track e genotipo
Track B: Q230P-class missense that is a proteasome substrate; a lysosomal route would make it ineffective; no meaning for null lesions

Profondità del beneficio cercato
deep only if stabilisation suffices; stabilisation without refolding is expected to raise inactive or insoluble protein for catalytic or folding lesions

Favorevole: premessa dimostrata
Cytosolic enzyme precedent (CBS): proteasome inhibition restored activity in vivo for a regulatory-domain allele and partly for catalytic-domain alleles; the route can be resolved per allele. Not WWOX.

Contraria: effetti opposti o controevidenza
Gain applies to wild type too (CBS wild-type half-life up about 7.7-fold); immune toxicity and toxic windows in controls; a heat-shock response confounds the readout; allele-selective failure (one allele not rescued in the mouse).

Mancante: premessa non dimostrata
That the Q230P-class protein is made and then cleared by the proteasome (the repository red-team record expects a lysosomal route); that recovered protein is functional; a safe exposure window.

Precedenti esterni (non sono evidenza WWOX)
CBS and ASPA precedents read at the passages the card cites (IT6 P6, 17 statements checked): boost without repair, insoluble stabilised protein (ASPA), protein up with activity flat (CBS with HSP70 or HSF1 co-expression), toxic windows including in controls.

Prossimo esperimento
Route-first fate experiment on the exact variant in an expression-matched WWOX-null human system: soluble, pellet and total protein, mRNA in the same wells, pulse-chase half-life, proteasome, lysosome and ERAD arms side by side each on the ruler line and the null line with a viability readout. No protein with synthesis present closes this card for the allele.

Stato dell'esperimento
PROPOSED, not run; no measurement of this kind exists in the repository; its absence does not block the computational work

AAV9-delivered human WWOX · AAV9-delivered human WWOX: endpoint families

other than spike-wave discharges CMP-03

PRIORITÀ DI RICERCA

AAV9-delivered human WWOX C01_aav9_wwox F2 sul bisogno Ripristinare la proteina WWOX nei neuroni aggiungendo il gene

AAV9-delivered human WWOX: endpoint families other than spike-wave discharges C04_aav9_endpoint_families F1 sul bisogno Ripristinare la proteina WWOX nei neuroni aggiungendo il gene C2 sul bisogno Scompenso metabolico sistemico

Piano di verifiche:

verificare un'ipotesi

Ripristinare la proteina WWOX nei neuroni aggiungendo il gene

Piano di verifiche:

chiarire una controevidenza

Scompenso metabolico sistemico

POSIZIONE DEL MODELLO

il modello non si applica a questa ipotesi (non valutata)

EVIDENZA OSSERVATA IN UN SISTEMA WWOX

Wwox-null mouse, vector given P0 to P5: spike-wave discharges not distinguishable from wild type at the high level (blinded, n 5); survival to 300 days at the high level comparable to the wild-type arm, the low level postpones death; weight equal at one age but survivor-conditioned; glucose unchanged; myelination dose relation never quantified; gliosis binary against wild type. Organoid AAV9-WWOX at week 10: hyperexcitability and transcript-level changes read at week 16 against non-treated organoids (not vector-treated).

▼ Dettagli: contesto, evidenza favorevole, contraria e mancante, prossimo esperimento

Track e genotipo
Track A: allele-blind in design (replaces the missing gene product); tested only in a null mouse; response in a missense or compound context with residual function is unmeasured

Profondità del beneficio cercato
deep: multi-endpoint restoration; shown for prevention in an early window, not for recovery after onset (one organoid endpoint aside)

Favorevole: premessa dimostrata
Prevention of the null phenotype by neuronal WWOX in the earliest postnatal window, in the strongest causal design of the corpus; neuron-restricted deletion reproduces the phenotype and neuron-restricted replacement reverses it.

Contraria: effetti opposti o controevidenza
Over-expression of a pro-apoptotic tumour suppressor with no ceiling measured and no surveillance; regionally uneven biodistribution; glucose endpoints null beside weight benefit (C2 on that need); wild-type arm itself lost about 29 percent by 300 days.

Mancante: premessa non dimostrata
Treatment after onset (mouse: unmeasured after P5); threshold versus steep continuum (two dose points); independent replication (two papers share a senior author); transfer to missense or compound alleles; tumour surveillance for an irreversibly delivered pro-apoptotic tumour suppressor; absolute vector quantity (uncertain by a factor of two in the source).

Precedenti esterni (non sono evidenza WWOX)
The two mouse papers and the organoid paper are the only sources; the survival and motor comparators were re-read at the source in IT6 P2; the organoid single-cell data are deposited but not accessible. No other gene-addition precedent is used.

Prossimo esperimento
An intermediate-level arm (a fraction of the high level) carried to 300 days with survival, weight, glucose, ECoG and quantified myelin in the same animals, plus a pre-treatment baseline and treated-after-onset arms with a same-age untreated comparator. A failure moves the card to C2 by rule; a treated-after-onset arm decides whether the word recovery may be used.

Stato dell'esperimento
PROPOSED, not run; no measurement of this kind exists in the repository; its absence does not block the computational work; needs a laboratory holding the vector, none known here

LDH inhibition or lowering CMP-04

PRIORITÀ DI RICERCA

LDH inhibition or lowering

E02_ldh_inhibition F3 sul bisogno Lattato e attività della lattato deidrogenasi P3 sul bisogno Respirazione mitocondriale e capacità di fosforilazione ossidativa

Piano di verifiche:

verificare un'ipotesi Lattato e attività della lattato deidrogenasi

POSIZIONE DEL MODELLO

nessun vantaggio dimostrato, esplorativo

EVIDENZA OSSERVATA IN UN SISTEMA WWOX

Transcript level only: LDHA higher in the knockout organoid and in shRNA progenitors (one library per arm); lactate and LDH activity higher in a human knockout fibroblast line; fly Ldh RNAi improved climbing once in an amyloid plus Wwox-reduction genotype (no Ldh-alone arm, no lactate readout). Human lactate in WWOX deficiency reported normal or extremely low in other sources. No compound has been given to any WWOX-deficient system on this need.

▼ Dettagli: contesto, evidenza favorevole, contraria e mancante, prossimo esperimento

Track e genotipo
Track A: allele-blind downstream metabolic lever; whether the axis is preserved in a catalytic-domain missense allele is an inference only

Profondità del beneficio cercato
partial: a metabolic consequence, not restoration of WWOX; no endpoint benefit is claimed

Favorevole: premessa dimostrata
LDHA transcript raised in two human neural transcript systems; HK and LDH activity raised in one human fibroblast line.

Contraria: effetti opposti o controevidenza

Stiripentol also inhibits mitochondrial complex I (isolated mouse heart and liver mitochondria, from tens of uM), so a metabolic effect is not an LDH effect and may recruit the integrated stress response; Ldh knockdown is detrimental in other fly amyloid models (authors' own citation); LDHB is the discordant gene in the P3 transfer test.

Mancante: premessa non dimostrata

LDH activity or lactate in WWOX-deficient human neural cells at matched maturity; that the elevation is not maturity or integrated-stress-response driven; that the candidate engages LDH in brain.

Precedenti esterni (non sono evidenza WWOX)

Rico-Molina 2024 read in part (inhibition data), Bacq 2024 review opened at the pharmacokinetic passage; Sada 2015 not lawfully acquirable (abstract and secondary descriptions only). Nedosiran shows LDHA lowering tolerated in liver.

Prossimo esperimento

Lactate (isotope or enantiomer resolved) and LDH activity, LDHA and LDHB separately, in WWOX-deficient versus isogenic human neural cells at matched maturity, with phospho-eIF2 alpha and ATF4 read beside LDH; the fly Ldh-alone arm; only then stiripentol against a selective LDHA tool at matched exposure. If lactate is not raised the need loses its WWOX observation and the card moves to P3 by rule.

Stato dell'esperimento

PROPOSED, not run; no measurement of this kind exists in the repository; its absence does not block the computational work

PDK inhibition [I01] · PDK inhibition [E03] CMP-05

PRIORITÀ DI RICERCA

PDK inhibition [I01]

I01_pdk_inhibition_dca_premise_seizure_glycaemia C3 sul bisogno Ingresso del piruvato nel ciclo di Krebs F3 sul bisogno Lattato e attività della lattato deidrogenasi C3 sul bisogno Controllo di crisi e spasmi come bisogno sintomatico

PDK inhibition [E03]

E03_pdk_inhibition_dca F3 sul bisogno Ingresso del piruvato nel ciclo di Krebs P3 sul bisogno Lattato e attività della lattato deidrogenasi P3 sul bisogno Controllo di crisi e spasmi come bisogno sintomatico

Piano di verifiche:

verificare un'ipotesi

Ingresso del piruvato nel ciclo di Krebs

Piano di verifiche:

verificare un'ipotesi Lattato e attività della lattato deidrogenasi

Piano di verifiche:

chiarire una controevidenza

Controllo di crisi e spasmi come bisogno sintomatico

Piano di verifiche:

chiarire una controevidenza

Scompenso metabolico sistemico

POSIZIONE DEL MODELLO

nessun vantaggio dimostrato, esplorativo

EVIDENZA OSSERVATA IN UN SISTEMA WWOX

PDK1 transcript up and HIF1alpha up in Wwox-null mouse embryonic fibroblasts; PDK transcript higher in a human knockout fibroblast line; PDK1 lower in sign only (inside the single-gene null) in two human neural transcript systems; PDH activity not different in the one human line; no PDH phosphorylation, flux or lactate measurement in any WWOX-deficient neural system. No compound given to any WWOX-deficient system.

▼ Dettagli: contesto, evidenza favorevole, contraria e mancante, prossimo esperimento

Track e genotipo
Track A: allele-blind in design; WWOX binds HIF1A through WW1, so the axis may be partly preserved in an SDR missense allele (inference)

Profondità del beneficio cercato
partial: a metabolic and possibly symptomatic effect; the card pursues no benefit endpoint and sharpens a decision

Favorevole: premessa dimostrata
The glycolytic end of the chain (LDHA, a HIF-type gene set) is raised in two human neural transcript systems; PDK-PDH restraint is documented after chemoconvulsant seizures in two non-WWOX epilepsy models.

Contraria: effetti opposti o controevidenza
Lowers glucose in adults and gluconeogenic precursors while hypoglycaemia is the Wwox-null phenotype (sign in a hypoglycaemic pup unmeasured); peripheral neuropathy is dose-limiting in adults; urinary oxalate +22 percent in a long-term cohort; rodent hepatotoxicity and neoplasia at high chronic doses; the general mechanism is self-inhibition of its own metabolism (GSTZ1), so exposure rises with repeated administration and varies with haplotype and age, whereas the one paediatric kinetic cohort (17 children) plateaued rather than accumulating - the two statements come from different sources and are not reconciled here; a slight pro-convulsant signal in one test.

Mancante: premessa non dimostrata
That PDH is restrained in a WWOX-deficient neural system and that the restraint is WWOX-dependent rather than seizure-secondary; the direction of the glucose response in a hypoglycaemic young animal.

Precedenti esterni (non sono evidenza WWOX)
Paediatric congenital lactic acidosis randomized trial (lactate moved, no clinical outcome in six months), a long-term follow-up, a MELAS trial stopped early for neuropathy; the one seizure-endpoint test in a PDH-restrained model was negative (abstract level, closed); several of these at abstract or review level only.

Prossimo esperimento
In WWOX-deficient human neural cells against matched controls: PDHA1 Ser293 phosphorylation with total PDH, PDH activity and a glucose tracer into acetyl-CoA, citrate and lactate, then with a PDK inhibitor against vehicle reading the genotype-by-inhibitor interaction (free DCA exposure and a second inhibitor of different chemistry as controls).

Stato dell'esperimento
PROPOSED, not run; no measurement of this kind exists in the repository; its absence does not block the computational work; cell access depends on a deposit that needs an access request that exists as a draft only and is not filed

Ketogenic and lactate-transport interventions · Ketogenic diet CMP-06

PRIORITÀ DI RICERCA

Ketogenic and lactate-transport interventions
E04_ketogenic_lactate_axis
P1 sul bisogno Ingresso del piruvato nel ciclo di Krebs

Ketogenic diet EB-KD-01
P3 sul bisogno Ingresso del piruvato nel ciclo di Krebs C2 sul bisogno Controllo di crisi e spasmi come bisogno sintomatico

Piano di verifiche:
verificare un'ipotesi
Ingresso del piruvato nel ciclo di Krebs

Piano di verifiche:
chiarire una controevidenza
Controllo di crisi e spasmi come bisogno sintomatico

POSIZIONE DEL MODELLO

Il modello non si applica a questa ipotesi (non valutata)

EVIDENZA OSSERVATA IN UN SISTEMA WWOX

One severe case lists the ketogenic diet among ineffective interventions; four sources report normal human metabolism in WWOX deficiency and one brain spectroscopy report has lactate extremely low. The fly lactate result is amyloid-conditional and cannot be used for or against the direction in WWOX-null neurons.

▼ Dettagli: contesto, evidenza favorevole, contraria e mancante, prossimo esperimento

Track e genotipo
Track A: allele-blind in design; mechanism for WWOX UNKNOWN

Profondità del beneficio cercato
partial: empirical seizure control; any bypass of a bioenergetic defect is hypothetical

Favorevole: premessa dimostrata
Ketogenic diet is an established empirical epilepsy treatment in general (card: empirical seizure-control use); nothing demonstrated for WWOX.

Contraria: effetti opposti o controevidenza
A ketone bypass would act without reactivating PDH, so it cannot test the PDK-PDH link; hypoglycaemia is the Wwox-null phenotype and a dietary intervention changes glucose supply (direction in this model UNKNOWN); human metabolic screens reported normal where done.

Mancante: premessa non dimostrata
Whether a WWOX-deficient human neural cell has a bioenergetic defect a ketone substrate would bypass; whether lactate is raised, normal or low in WWOX-deficient brain across ages.

Precedenti esterni (non sono evidenza WWOX)
General epilepsy use; no precedent was re-read for this comparison and the card records three earlier depths, not repeated.

Prossimo esperimento
PDH-specific readouts in WWOX-deficient human neural cells or the existing knockout organoids against isogenic controls: PDH activity in a phosphorylation-preserving lysate, phospho-PDHA1 Ser293, 13C-glucose or ketone tracing, pyruvate-to-lactate ratio, NADH/NAD+. Seahorse alone is a screen and not a PDH measure.

Stato dell'esperimento
PROPOSED, not run; no measurement of this kind exists in the repository; its absence does not block the computational work

NMDAR intervention class for the neuron-specific Wwox bursting phenotype CMP-07

PRIORITÀ DI RICERCA

NMDAR intervention class for the neuron-specific Wwox bursting phenotype

F04_nmdar_intervention_class

F3 sul bisogno Guadagno eccitatorio di rete e scariche a raffica

Piano di verifiche:

verificare un'ipotesi

Guadagno eccitatorio di rete e scariche a raffica

POSIZIONE DEL MODELLO

il modello non si applica a questa ipotesi (non valutata)

EVIDENZA OSSERVATA IN UN SISTEMA WWOX

Neuron-specific knockout cortical slice (single paper): d-APV abolishes bursting (5 slices from 3 animals, no significance marker in the figure) and the delayed evoked response; sEPSC about 6 percent higher; sIPSC amplitude 27.5 against 57.3 pA (the larger change is inhibitory); hippocampal slices do not burst. No memantine, ketamine, GluN2B modulator or AMPAR antagonist was applied in a WWOX system.

▼ Dettagli: contesto, evidenza favorevole, contraria e mancante, prossimo esperimento

Track e genotipo
Track A: acts on a network state whatever the allele; the one electrophysiology source is a neuron-specific knockout, so missense lesions are untested

Profondità del beneficio cercato
partial: reduction of a network state (bursting); the card says no developmental benefit is implied

Favorevole: premessa dimostrata
Bursting needs NMDAR activity in this slice; the knockout rests more depolarised than wild type.

Contraria: effetti opposti o controevidenza
Selfotel failed with a harm signal in adults; NMDAR blockade in the developing rat brain triggers apoptosis (abstract level); memantine accelerated seizure stages in an adult rat oxygen model; perampanel carries a boxed warning for psychiatric and behavioural reactions; sustained NMDAR dampening may affect activity-dependent myelination (F02) in a developing brain; AMPAR block can abolish bursts without discriminating the NMDAR sign.

Mancante: premessa non dimostrata
NMDAR gain (versus permissive dependence on depolarisation and weak inhibition); the fraction of NMDAR current that must be blocked; the recording-bath Mg2+; any subunit data; that the finding extends beyond cortex slice or to other alleles.

Precedenti esterni (non sono evidenza WWOX)

GluN2B negative modulators effective at more than 80 percent of the 27 tested gain-of-function GRIN variants in vitro (abstract only; source form); memantine improvement in any domain in a GRIN series 14 of 19 gain-of-function against 3 of 15 other (partial read; seizure-frequency difference not significant); selfotel stopped in adult stroke and head injury trials. All in gain-of-function disease, where the sign is known.

Prossimo esperimento

Step E1 in the source preparation (neuron-specific knockout, littermates, P13 to P17, layer II/III): isolated NMDAR current under voltage control with AMPAR and GABA-A blocked, amplitude, decay, NMDA:AMPA ratio, GluN2B and GluN2A shares, resting potential; then d-APV and a GluN2B tool concentration-response on the isolated current, the delayed response and bursts; a pore blocker only afterwards as a translation probe with Mg2+ and holding potential stated.

Stato dell'esperimento

PROPOSED, not run; no measurement of this kind exists in the repository; its absence does not block the computational work; the knockout line is not held here

Extracellular magnesium as the physiological voltage-dependent NMDAR gate CMP-08

PRIORITÀ DI RICERCA

Extracellular magnesium as the physiological voltage-dependent NMDAR gate

P11_magnesium_nmdar_gate

P3 sul bisogno Guadagno eccitatorio di rete e scariche a raffica

Piano di verifiche:

verificare un'ipotesi

Guadagno eccitatorio di rete e scariche a raffica

POSIZIONE DEL MODELLO

il modello non si applica a questa ipotesi (non valutata)

EVIDENZA OSSERVATA IN UN SISTEMA WWOX

None for magnesium: no WWOX measurement of Mg2+, NMDAR current or block was found (zero hits for WWOX with magnesium in the searches run). The convergent WWOX observations are the d-APV result, a resting potential of -73.62 mV in knockout against -79.25 mV in wild type, and halved inhibitory current amplitude.

▼ Dettagli: contesto, evidenza favorevole, contraria e mancante, prossimo esperimento

Track e genotipo

Track A: allele-blind; Track B unmeasured

Profondità del beneficio cercato

partial at best: a reduction of a network state; no developmental benefit implied

Favorevole: premessa dimostrata

The knockout slice bursts in a medium whose incubation solution contains 1.3 mM Mg2+ (so the bursting is probably not a magnesium-free artefact; the recording bath is not stated, which is an inference).

Contraria: effetti opposti o controevidenza

Sustained NMDAR dampening may affect activity-dependent myelination or synaptic maturation in a developing brain (unmeasured); a graded rise in an ion that is already present removes only part of the residual conductance while d-APV removes all; one case on vitamin B6 and zonisamide without relief is recorded as counter-data for a related nutrient strand.

Mancante: premessa non dimostrata

That the burst depends on the Mg2+ block; that a reachable Mg2+ rise lowers bursts without altering wild-type activity; that any preparation raises brain extracellular free Mg2+ in a child; serum and CSF Mg in WWOX-DEE.

Precedenti esterni (non sono evidenza WWOX)

Infantile spasms: intravenous magnesium sulfate with ACTH outperformed ACTH alone in 38 infants (abstract only); an open-label trial with an oral phase (48 analysed, both arms on ACTH and antiseizure drugs) found no difference (p 0.69). One mouse study read: oral magnesium sulfate raised serum Mg and left CSF Mg unchanged; the threonate raised CSF Mg by about a third in healthy mice (six per subgroup).

Prossimo esperimento

Graded bath Mg2+ around the slice value, in both directions, in knockout, heterozygote and wild-type neocortical slices at P13 to P17 in the source medium, then d-APV as ceiling; endpoints burst incidence, frequency, duration and amplitude, plus NMDA:AMPA ratio; in parallel total and ionised Mg in serum and CSF of knockout against control mice and the CSF response to one oral preparation. A rise that cannot lower bursts inside a reachable range closes the strand.

Stato dell'esperimento

PROPOSED, not run; no measurement of this kind exists in the repository; its absence does not block the computational work; slice electrophysiology line not held here

Glial-reactivity modulators and the unresolved sign and order of gliosis in … CMP-09

PRIORITÀ DI RICERCA

Glial-reactivity modulators and the unresolved sign and order of gliosis in …

J01_glial_reactivity_modulation

P1 sul bisogno

Neuroinfiammazione e gliosi

Piano di verifiche:

risolvere una premessa

Neuroinfiammazione e gliosi

POSIZIONE DEL MODELLO

il modello non si applica a questa ipotesi (non valutata)

EVIDENZA OSSERVATA IN UN SISTEMA WWOX

Gliosis raised in the null mouse and the P47T homozygote and returned toward baseline by neuron-restricted replacement (GFAP-positive cells about 44 to about 5); the opposite sign in a Wwox-mutant rat cortex (about half the GFAP and Iba1 cells from PND 5); the human organoid shows no reactive programme and has no microglia. No modulator given to any WWOX system.

▼ Dettagli: contesto, evidenza favorevole, contraria e mancante, prossimo esperimento

Track e genotipo

Track A: allele-blind in design; no glial datum exists in a catalytic-domain missense allele

Profondità del beneficio cercato

partial: protection or symptom relief downstream of neuronal loss; the causal compartment is neuronal, so a glial drug is not expected to touch the cause

Favorevole: premessa dimostrata

Glial change accompanies the phenotype in several mouse models and tracks rescue; a neuron-restricted deletion also shows astrocyte-activation-type genes, so glial change occurs without an astrocyte-intrinsic lesion.

Contraria: effetti opposti o controevidenza

If glial reactivity is protective or maturational in a developing Wwox-deficient brain (the rat sign), suppression could harm; the same neuron-restricted rescue that lowers GFAP also stops seizures, so the data cannot assign the fall to a glial route; long-term disability persists after acute control in the human precedents.

Mancante: premessa non dimostrata

The sign of the glial difference before the first spike; whether glia are upstream of any harm; species versus allele versus region versus age as the source of the sign conflict; any astrocyte-specific deletion result.

Precedenti esterni (non sono evidenza WWOX)

IL-1 axis agents in febrile-infection or cryptogenic refractory epilepsy: acute seizure-control reports in uncontrolled reviews (up to 73 percent within two weeks in 75 pooled patients), no demonstrated long-term benefit in a 14-patient series; minocycline in three patients uninformative; a caspase-1 inhibitor phase II result unverifiable at trial level and without effect in a rat infantile-spasms model. Abstract or review level.

Prossimo esperimento

Wwox-null or neuron-restricted deletion pups with wild-type littermates: GFAP and Iba1 density and area fraction in cortex and hippocampus at P5, P8, P11, P14 and P17 on the same sections, with continuous cortical recording from the earliest feasible age, each animal classed before or after its first spike, vector-treated controls included. The sign before the first spike decides whether glia are a readout, a seizure consequence or a lever. No compound arm is designed.

Stato dell'esperimento

PROPOSED, not run; no measurement of this kind exists in the repository; its absence does not block the computational work; recording from P5 is a feasibility risk